Kubernetes & Rancher · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Core objects

- Pod (smallest unit), Deployment (replicas, rolling updates), ReplicaSet.
- Service: ClusterIP, NodePort, LoadBalancer. Ingress: HTTP routing rules.
- ConfigMap (config), Secret (base64, enable encryption at rest), PVC (storage).
- StatefulSet (stable identity), DaemonSet (one pod per node), Job and CronJob.
- Namespaces, ResourceQuota, RBAC (Role and RoleBinding), NetworkPolicy.

Control plane and nodes

- API server, etcd (cluster state, back it up), scheduler, controller manager.
- kubelet on each node runs pods; kube-proxy handles service networking.
- NotReady node: check kubelet, network and resources.

kubectl essentials

- get pods -A, describe pod, logs --previous, exec -it pod -- sh.
- apply -f, rollout status, rollout history, rollout undo, rollout restart.
- scale --replicas=5, top nodes and top pods (needs metrics-server).
- get endpoints svc to check that a service has ready pods.

Common statuses

- Pending: cannot schedule (resources, volumes, taints).
- ContainerCreating stuck: volume, image or network plugin errors.
- CrashLoopBackOff: app keeps crashing; check logs --previous.
- ImagePullBackOff: wrong image or missing imagePullSecret.
- OOMKilled: memory limit exceeded.

Scheduling and resources

- Requests are for scheduling; limits cap usage (memory over limit = OOMKilled).
- Taints and tolerations keep pods off nodes; node affinity places them.
- HPA scales pods; Cluster Autoscaler scales nodes.
- Probes: liveness restarts, readiness controls traffic.

Maintenance and upgrades

- cordon (no new pods), drain (evict pods), PodDisruptionBudget keeps replicas up.
- Upgrade order: back up etcd, upgrade control plane, then nodes one by one, then validate.
- Rancher manages many clusters; RKE2 (secure) and K3s (lightweight) distributions.
- Helm charts and Rancher Apps install monitoring and ingress controllers.

Troubleshooting

- Service unreachable: selectors, endpoints, readiness, network policies.
- Ingress 404: host, path, pathType, backend service and port, ingress class.
- DNS: nslookup service.namespace.svc.cluster.local from a test pod; check CoreDNS.
- Check your kubectl context before running commands.